TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 5

5-MINUTE REVISION

Central Tendency II

From the chapter's own revision points, formulae and questions.

Selected from Chapter 5 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Geometric mean → suitable for growth rates, ratios, percentages, and multiplicative changes.
- Harmonic mean → suitable for rates, ratios, and reciprocal relationships.
- Geometric mean → nth root of the product of n observations.
- Harmonic mean → reciprocal of the arithmetic mean of reciprocals.
- Geometric mean gives greater weight to smaller values and less weight to larger values.

Quick revision · 2

- Harmonic mean gives greater weight to smaller values.
- The logarithmic GM requires positive observations; the product GM of non-negative values is zero if any value is zero.
- The standard statistical GM is generally unsuitable for negative observations.
- Arithmetic mean, geometric mean, and harmonic mean are called Pythagorean means.
- Mathematical averages → arithmetic mean, geometric mean, and harmonic mean.

Quick revision · 3

- Positional measures → median, quartiles, percentiles, and deciles; mode is based on frequency.
- Quartiles → divide data into 4 equal parts.
- Q_1 → 25th percentile.
- Q_2 → 50th percentile → Median.
- Q_3 → 75th percentile.

Quick revision · 4

- Percentiles → divide data into 100 equal parts.
- $P_{25} = Q_1$.
- $P_{50} = Q_2 = \text{Median}$.
- $P_{75} = Q_3$.
- Deciles → divide data into 10 equal parts.

Quick revision · 5

- $D_5 = Q_2 = P_{50} = \text{Median}$.
- For raw data, arrange observations in ascending order before finding positional measures.
- For discrete frequency data, use cumulative frequencies to locate the required position.
- For grouped frequency data, first identify the relevant quartile, percentile, or decile class.
- In grouped data, interpolation is then used to calculate the required value.

Important formulae · 1

Geometric mean:

$$GM = \sqrt[n]{\prod_{i=1}^{n} x_i}$$

Geometric mean using logarithms:

$$\log GM = \frac{\sum_{i=1}^{n} \log x_i}{n}$$

Geometric mean using antilogarithms:

$$GM = \text{Antilog}\left(\frac{\sum_{i=1}^{n} \log x_i}{n}\right)$$

Important formulae · 2

Geometric mean for a frequency distribution:

$$GM = \text{Antilog}\left(\frac{\sum_{i=1}^{k} f_i \log x_i}{n}\right)$$

Harmonic mean:

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Harmonic mean for a frequency distribution:

$$HM = \frac{n}{\sum_{i=1}^{k} f_i \frac{1}{x_i}}$$

Important formulae · 3

First quartile for raw data:

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

Third quartile for raw data:

$$Q_3 = \left(\frac{3(n+1)}{4}\right)^{\text{th}} \text{ item}$$

First quartile for grouped data:

$$Q_1 = l_1 + \frac{\frac{n}{4} - m_1}{f_1} \times c_1$$

Important formulae · 4

Third quartile for grouped data:

$$Q_3 = l_3 + \frac{\frac{3n}{4} - m_3}{f_3} \times c_3$$

Percentile for raw data:

$$P_p = \left(\frac{p(n+1)}{100}\right)^{\text{th}} \text{ item}$$

Percentile for grouped data:

$$P_p = l + \frac{\left(\frac{pn}{100}\right) - cf}{f} \times c$$

Important formulae · 5

Decile for raw data:

$$D_d = \left(\frac{d(n+1)}{10}\right)^{\text{th}} \text{ item}$$

Decile for grouped data:

$$D_d = l + \frac{\left(\frac{dn}{10}\right) - cf}{f} \times c$$

Relationship among Pythagorean means (the first equality is for two positive observations):

$$AM \times HM = GM^2$$

$$AM \geq GM \geq HM$$

Worked example from the chapter

A car travels a certain distance from City A to City B at a speed of 60 km/h, and returns the same distance from City B to City A at a speed of 90 km/h. What is the average speed for the entire trip?

Solution

To find the average speed when traveling the same distance at two different speeds, we use the harmonic mean. The harmonic mean for the speed is $\frac{2 \cdot S_1 \cdot S_2}{S_1 + S_2}$, where $S_1 = 60\,\text{km/h}$ is the speed from City A to City B, $S_2 = 90\,\text{km/h}$ is the speed from City B to City A.

In this case, we are calculating the average speed over the entire round trip. The harmonic mean is used because it accounts for the fact that traveling at different speeds over the same distance results in an overall average speed that is closer to the lower of the two speeds, rather than simply averaging the two speeds.

Now, applying the harmonic mean formula:

$S_{\text{avg}} = \frac{2 \cdot 60 \cdot 90}{60+90} = \frac{10800}{150} = 72\,\text{km/h}$

So, the average speed for the entire trip is 72 km/h.

Self-check from the chapter

1. For positive observations, the nth root of their product is the _____ mean. **[Geometric]**
2. The reciprocal of the arithmetic mean of reciprocals is the _____ mean. **[Harmonic]**
3. The mean suitable for averaging positive growth factors is the _____ mean. **[Geometric]**
4. The mean suitable for averaging speeds over equal distances is the _____ mean. **[Harmonic]**